

# Midterm Exam Solutions

**Statistical Methods(통계적방법론)**

**2026 1st semester**

|                                                                                 |
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| Section(교반): <u>A1</u> Cadet Number(교번): _____ Name(성명): _____ Score(점수): _____ |
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- All solutions must include a detailed step-by-step explanation.

1. [50 points] Consider the multiple linear regression model:

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \eta^{-1}) \quad \text{independently for } i = 1, \dots, n, \quad (1)$$

where  $\mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\beta} \in \mathbb{R}^p$  and  $X = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top \in \mathbb{R}^{n \times p}$  is full rank. ( $\text{rank}(X) = p$  with  $n > p$ ) Letting  $\mathbf{y} = (y_1, \dots, y_n)^\top$  and  $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)^\top$ , (1) can equivalently be written as

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim MVN(\mathbf{0}, \eta^{-1}I_n).$$

The log-likelihood of  $\boldsymbol{\beta}$  and  $\eta$  is

$$\ell(\boldsymbol{\beta}, \eta) = -\frac{n}{2} \log(2\pi) + \frac{n}{2} \log \eta - \frac{\eta}{2} (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}).$$

(1) Show that the score function of  $\boldsymbol{\beta}$  and  $\eta$  is

$$S(\boldsymbol{\beta}, \eta) = \begin{pmatrix} \frac{\partial \ell}{\partial \boldsymbol{\beta}} \\ \frac{\partial \ell}{\partial \eta} \end{pmatrix} = \begin{pmatrix} \eta X^\top (\mathbf{y} - X\boldsymbol{\beta}) \\ \frac{n}{2\eta} - \frac{1}{2} (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}) \end{pmatrix}.$$

Solution: Using  $\frac{\partial}{\partial \boldsymbol{\beta}} (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}) = -2X^\top (\mathbf{y} - X\boldsymbol{\beta})$ ,

$$\frac{\partial \ell}{\partial \boldsymbol{\beta}} = -\frac{\eta}{2} \cdot (-2X^\top (\mathbf{y} - X\boldsymbol{\beta})) = \eta X^\top (\mathbf{y} - X\boldsymbol{\beta}).$$

Differentiating with respect to  $\eta$ :

$$\frac{\partial \ell}{\partial \eta} = \frac{n}{2\eta} - \frac{1}{2} (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}).$$

Combining the two partial derivatives gives  $S(\boldsymbol{\beta}, \eta)$  as stated.  $\square$

(2) Show that the maximum likelihood estimator(MLE) that solves  $S(\boldsymbol{\beta}, \eta) = 0$  is

$$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}, \quad \hat{\eta} = \frac{n}{(\mathbf{y} - X\hat{\boldsymbol{\beta}})^\top (\mathbf{y} - X\hat{\boldsymbol{\beta}})}.$$

(Hint: find  $\hat{\boldsymbol{\beta}}$  first.)

Solution: Setting  $\frac{\partial \ell}{\partial \boldsymbol{\beta}} = \mathbf{0}$  and dividing by  $\eta > 0$  gives the normal equations  $X^\top X\boldsymbol{\beta} = X^\top \mathbf{y}$ . Since

$\text{rank}(X) = p$  makes  $X^\top X$  invertible,  $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$ . Substituting  $\hat{\boldsymbol{\beta}}$  into  $\frac{\partial \ell}{\partial \eta} = 0$  gives

$$\frac{n}{2\hat{\eta}} = \frac{1}{2} (\mathbf{y} - X\hat{\boldsymbol{\beta}})^\top (\mathbf{y} - X\hat{\boldsymbol{\beta}}), \text{ hence } \hat{\eta} = \frac{n}{(\mathbf{y} - X\hat{\boldsymbol{\beta}})^\top (\mathbf{y} - X\hat{\boldsymbol{\beta}})}. \quad \square$$

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(3) Show that the Fisher information matrix is

$$\mathcal{I}(\beta, \eta) = -E \begin{bmatrix} \frac{\partial^2 \ell}{\partial \beta \partial \beta^\top} & \frac{\partial^2 \ell}{\partial \beta \partial \eta} \\ \frac{\partial^2 \ell}{\partial \eta \partial \beta^\top} & \frac{\partial^2 \ell}{\partial \eta^2} \end{bmatrix} = \begin{bmatrix} \eta X^\top X & \mathbf{0} \\ \mathbf{0}^\top & \frac{n}{2\eta^2} \end{bmatrix}.$$

Solution:

We compute each block of second derivatives.

Block  $(\beta, \beta)$ :

$$\frac{\partial^2 \ell}{\partial \beta \partial \beta^\top} = \frac{\partial}{\partial \beta^\top} [\eta X^\top (\mathbf{y} - X\beta)] = -\eta X^\top X.$$

$$\text{Hence } -E \left[ \frac{\partial^2 \ell}{\partial \beta \partial \beta^\top} \right] = \eta X^\top X.$$

Block  $(\beta, \eta)$ :

$$\frac{\partial^2 \ell}{\partial \beta \partial \eta} = \frac{\partial}{\partial \eta} [\eta X^\top (\mathbf{y} - X\beta)] = X^\top (\mathbf{y} - X\beta).$$

$$\text{Taking expectation: } E[X^\top (\mathbf{y} - X\beta)] = X^\top (X\beta - X\beta) = \mathbf{0}, \text{ so } -E \left[ \frac{\partial^2 \ell}{\partial \beta \partial \eta} \right] = \mathbf{0}.$$

Block  $(\eta, \eta)$ :

$$\frac{\partial^2 \ell}{\partial \eta^2} = -\frac{n}{2\eta^2}.$$

$$\text{Hence } -E \left[ \frac{\partial^2 \ell}{\partial \eta^2} \right] = \frac{n}{2\eta^2}.$$

Assembling the blocks:

$$\mathcal{I}(\beta, \eta) = \begin{bmatrix} \eta X^\top X & \mathbf{0} \\ \mathbf{0}^\top & \frac{n}{2\eta^2} \end{bmatrix}. \quad \square$$

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(4) Using the Fisher information in problem (3), find the asymptotic distribution of  $\hat{\beta}$ .

Solution:

By general MLE theory, the MLE  $(\hat{\beta}, \hat{\eta})$  is asymptotically normal with covariance  $\mathcal{I}(\beta, \eta)^{-1}$ . The  $(\beta, \beta)$ -block of the Fisher information is  $\mathcal{I}_{\beta\beta} = \eta X^\top X$ , and because the off-diagonal blocks are zero, its inverse is simply  $\mathcal{I}_{\beta\beta}^{-1} = \eta^{-1}(X^\top X)^{-1}$ . Therefore,

$$\hat{\beta} \xrightarrow{d} MVN(\beta, \eta^{-1}(X^\top X)^{-1}).$$

(In fact, for the normal linear model this result holds exactly in finite samples, not just asymptotically.)

(5) Consider the hypothesis test:

$$H_0 : \beta = \mathbf{0}, \quad H_1 : \beta \neq \mathbf{0}.$$

Show that the Wald test statistic is

$$W = (\hat{\beta}^\top \quad 0) \mathcal{I}(\hat{\beta}, \hat{\eta}) \begin{pmatrix} \hat{\beta} \\ 0 \end{pmatrix} = \frac{(X\hat{\beta})^\top (X\hat{\beta})}{(\mathbf{y} - X\hat{\beta})^\top (\mathbf{y} - X\hat{\beta})/n}.$$

What is the (asymptotic) null distribution of the test statistic?

Solution:

*Derivation of  $W$ .* Evaluating the Fisher information at  $(\hat{\beta}, \hat{\eta})$  gives

$$\mathcal{I}(\hat{\beta}, \hat{\eta}) = \begin{bmatrix} \hat{\eta} X^\top X & \mathbf{0} \\ \mathbf{0}^\top & \frac{n}{2\hat{\eta}^2} \end{bmatrix}.$$

Since the off-diagonal blocks are zero, only the  $(\beta, \beta)$ -block contributes:

$$W = (\hat{\beta}^\top \quad 0) \mathcal{I}(\hat{\beta}, \hat{\eta}) \begin{pmatrix} \hat{\beta} \\ 0 \end{pmatrix} = \hat{\beta}^\top (\hat{\eta} X^\top X) \hat{\beta} = \hat{\eta} (X\hat{\beta})^\top (X\hat{\beta}).$$

Substituting  $\hat{\eta} = \frac{n}{(\mathbf{y} - X\hat{\beta})^\top (\mathbf{y} - X\hat{\beta})}$ :

$$W = \frac{n (X\hat{\beta})^\top (X\hat{\beta})}{(\mathbf{y} - X\hat{\beta})^\top (\mathbf{y} - X\hat{\beta})} = \frac{(X\hat{\beta})^\top (X\hat{\beta})}{(\mathbf{y} - X\hat{\beta})^\top (\mathbf{y} - X\hat{\beta})/n}. \quad \square$$

*Null distribution.* Under  $H_0 : \beta = \mathbf{0}$ , by the asymptotic normality of  $\hat{\beta}$  from part (4),  $\hat{\beta}$  has  $p$  components, so

$$W \xrightarrow{d} \chi^2(p).$$

That is, the Wald statistic follows asymptotically a chi-squared distribution with  $p$  degrees of freedom under the null hypothesis.

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2. [30 points] Consider the logistic regression model with a single binary covariate  $x_i \in \{0, 1\}$ :

$$\log \frac{p_i}{1 - p_i} = \beta_0 + \beta_1 x_i, \quad y_i \sim \text{Bernoulli}(p_i) \text{ independently, } i = 1, \dots, n.$$

The data are summarized in the following contingency table:

|   |       | Y                 |                   |                                     |
|---|-------|-------------------|-------------------|-------------------------------------|
|   |       | 0                 | 1                 | Total                               |
| X | 0     | $n_{00}$          | $n_{01}$          | $n_{00} + n_{01} = n_0$             |
|   | 1     | $n_{10}$          | $n_{11}$          | $n_{10} + n_{11} = n_1$             |
|   | Total | $n_{00} + n_{10}$ | $n_{01} + n_{11}$ | $n_{00} + n_{01} + n_{10} + n_{11}$ |

where  $n_{00}, n_{01}, n_{10}, n_{11} > 0$ .

(1) Show that

$$p_i = \frac{1}{1 + \exp(-\beta_0 - \beta_1 x_i)}.$$

Solution: Exponentiating both sides of the logit equation gives  $\frac{p_i}{1 - p_i} = e^{\beta_0 + \beta_1 x_i}$ , so  $p_i = (1 - p_i)e^{\beta_0 + \beta_1 x_i}$ .

Solving for  $p_i$ :

$$p_i = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} = \boxed{\frac{1}{1 + \exp(-\beta_0 - \beta_1 x_i)}}.$$

(2) For subjects with  $x_i = 0$ ,  $p_i = p(0) = \frac{e^{\beta_0}}{1 + e^{\beta_0}}$ ; for  $x_i = 1$ ,  $p_i = p(1) = \frac{e^{\beta_0 + \beta_1}}{1 + e^{\beta_0 + \beta_1}}$ . Express the odds ratio

(OR)

$$\text{OR} = \frac{p(1)/(1 - p(1))}{p(0)/(1 - p(0))}$$

in terms of  $\beta_0$  and  $\beta_1$ .

Solution: From the logistic model, the odds for each group are

$$\frac{p(0)}{1 - p(0)} = e^{\beta_0}, \quad \frac{p(1)}{1 - p(1)} = e^{\beta_0 + \beta_1}.$$

Therefore,

$$\text{OR} = \frac{p(1)/(1 - p(1))}{p(0)/(1 - p(0))} = \frac{e^{\beta_0 + \beta_1}}{e^{\beta_0}} = \boxed{e^{\beta_1}}.$$

Hence  $\beta_1 = \log(\text{OR})$  is the log odds ratio of  $Y = 1$  for  $X = 1$  versus  $X = 0$ .

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(3) The MLEs of  $p(0)$  and  $p(1)$  are given as

$$\hat{p}(0) = \frac{n_{01}}{n_{00} + n_{01}}, \quad \hat{p}(1) = \frac{n_{11}}{n_{10} + n_{11}}.$$

Show that the maximum likelihood estimator (MLE) of log odds ratio (OR) is

$$\log \widehat{\text{OR}} = \log \frac{\hat{p}(1)/(1 - \hat{p}(1))}{\hat{p}(0)/(1 - \hat{p}(0))} = \log \frac{n_{11} n_{00}}{n_{10} n_{01}}.$$

Solution:

By the invariance property of MLEs, the MLE of any function  $g(\theta)$  is  $g(\hat{\theta})$ . From part (2),  $\text{OR} = e^{\beta_1}$ , so  $\log \text{OR} = \beta_1$ , and the MLE is obtained by plugging in  $\hat{p}(0)$  and  $\hat{p}(1)$ .

First, compute each estimated odds. Since  $\hat{p}(0) = \frac{n_{01}}{n_0}$ , we have  $1 - \hat{p}(0) = \frac{n_{00}}{n_0}$ , so

$$\frac{\hat{p}(0)}{1 - \hat{p}(0)} = \frac{n_{01}}{n_{00}}.$$

Similarly,  $\hat{p}(1) = \frac{n_{11}}{n_1}$  gives  $1 - \hat{p}(1) = \frac{n_{10}}{n_1}$ , so

$$\frac{\hat{p}(1)}{1 - \hat{p}(1)} = \frac{n_{11}}{n_{10}}.$$

Therefore, the estimated odds ratio is

$$\widehat{\text{OR}} = \frac{\hat{p}(1)/(1 - \hat{p}(1))}{\hat{p}(0)/(1 - \hat{p}(0))} = \frac{n_{11}/n_{10}}{n_{01}/n_{00}} = \frac{n_{11} n_{00}}{n_{10} n_{01}},$$

and taking the logarithm gives

$$\log \widehat{\text{OR}} = \log \frac{n_{11} n_{00}}{n_{10} n_{01}}. \quad \square$$

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3. [20 points] Consider the multiple linear regression model in problem 1.

(1) Find the ridge estimator of  $\beta$ , denoted  $\hat{\beta}_\lambda$ , that minimizes

$$\ell(\beta) = (\mathbf{y} - X\beta)^\top (\mathbf{y} - X\beta) + \lambda \beta^\top \beta = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2 + \lambda \beta^\top \beta.$$

where  $\lambda \geq 0$ .

Solution: Differentiate  $\ell(\beta)$  with respect to  $\beta$  and set to zero:

$$\frac{\partial \ell}{\partial \beta} = -2X^\top (\mathbf{y} - X\beta) + 2\lambda \beta = \mathbf{0}.$$

Rearranging gives the normal equations for the ridge estimator:

$$(X^\top X + \lambda I_p) \beta = X^\top \mathbf{y}.$$

Since  $X^\top X + \lambda I_p$  is positive definite (hence invertible) for any  $\lambda > 0$ ,

$$\boxed{\hat{\beta}_\lambda = (X^\top X + \lambda I_p)^{-1} X^\top \mathbf{y}.} \quad (2)$$

(2) Find  $E(\hat{\beta}_\lambda)$ . What is the value of  $\lambda$  that makes  $E(\hat{\beta}_\lambda) = \beta$ ?

Solution: Substituting  $\mathbf{y} = X\beta + \varepsilon$  into (2) and taking expectation:

$$\begin{aligned} E(\hat{\beta}_\lambda) &= (X^\top X + \lambda I_p)^{-1} X^\top E(\mathbf{y}) \\ &= (X^\top X + \lambda I_p)^{-1} X^\top X \beta. \end{aligned} \quad (3)$$

In general,  $E(\hat{\beta}_\lambda) \neq \beta$  unless  $\lambda = 0$ , in which case

$$(X^\top X + 0 \cdot I_p)^{-1} X^\top X = (X^\top X)^{-1} X^\top X = I_p,$$

so  $E(\hat{\beta}_0) = \beta$ . Hence,  $\hat{\beta}_\lambda$  is biased for  $\lambda > 0$ , and the value  $\boxed{\lambda = 0}$  makes the estimator unbiased.